

WEEK 2 REPORT

Multimodal Baseline Model Development — BLIP Image Captioning on Flickr8k

1. Objective

Week 2 established a multimodal image-captioning baseline on Flickr8k. The work covered zero-shot BLIP inference, quantitative evaluation (BLEU, ROUGE, METEOR), fine-tuning experiments with MLflow tracking, and selection of a final baseline configuration for deployment.

2. Model: BLIP

BLIP uses a **Vision Transformer (ViT)** encoder for images and a **Transformer text decoder** for caption generation via cross-attention. Two decoding strategies were compared: **greedy** (pick best word each step) and **beam search** (5 beams, pick globally best sentence). All captions limited to 30 tokens.

3. Zero-Shot Baseline Results

Pretrained BLIP evaluated zero-shot on 300 validation images (seed 42):

Metric	BLEU	ROUGE-1	ROUGE-L	METEOR
Zero-shot	20.37	50.73	47.77	36.88

ROUGE-1 $\approx$ 51 shows scene understanding; low BLEU indicates phrasing mismatch — the gap fine-tuning addresses.

4. Fine-Tuning Experiments

Vision encoder **frozen**; only text decoder fine-tuned. Three experiments on Colab T4 GPU (2 epochs, batch 8, 150-image eval):

Run	LR	Train	Loss	Greedy BLEU	Greedy METEOR	Beam BLEU	Beam METEOR
exp1_baseline	5e-5	200	2.49	21.78	45.09	23.59	46.55
exp2_high_lr	1e-4	200	2.45	18.89	45.24	23.76	49.42
exp3_large_subset	5e-5	500	2.29	21.68	46.78	22.87	47.58

- **More data > higher LR:** exp3 (500 images) had lowest loss (2.29) and best greedy METEOR (46.78).
- **LR 5e-5 more stable:** exp2 (1e-4) showed greedy–beam inconsistency (BLEU 18.89 vs 23.76), a sign of aggressive updates.
- **Beam > greedy:** beam search improved BLEU by 1–5 pts across all runs.
- **+10 METEOR gain:** fine-tuning lifted METEOR from 36.88 (zero-shot) to 45–49, confirming adaptation to Flickr8k style.

5. Chosen Baseline

exp3_large_subset (lr 5e-5, 500 images, beam decoding) — chosen for overall consistency, lowest training loss, and strongest greedy-mode performance. Configuration saved in `best_config.json`.

6. MLflow Experiment Dashboard

All 5 runs tracked in a single MLflow experiment (BLIP-Flickr8k-Captioning) with SQLite backend:

Run details

Run ID:	07defe96a9174f72b6bf2c2...	f468535adc2a40b586a8cd...	ef83d5cc039344bc98498a...	e72f1446dc154123bc19b9...	2c1
Run Name:	exp3_large_subset	exp1_baseline	blip_text_decoder_finetune	zero_shot_blip_baseline	exp
Start Time:	07/20/2026, 11:23:59 PM	07/20/2026, 11:19:12 PM	07/16/2026, 04:47:01 PM	07/16/2026, 04:46:42 PM	07/
End Time:	07/20/2026, 11:26:37 PM	07/20/2026, 11:21:35 PM	07/16/2026, 04:58:42 PM	07/16/2026, 04:46:42 PM	07/
Duration:	2.6min	2.4min	11.7min	411ms	2.3i

Figure 1: Run details — 5 runs with timestamps and durations

beam_BLEU	22.87	23.59	NaN	NaN	2
beam_METEOR	47.58	46.55	NaN	NaN	4
beam_ROUGE_L	53.59	53.45	NaN	NaN	5
beam_ROUGE_L	50.35	50.67	NaN	NaN	5
bleu	NaN	NaN	NaN	20.37	1
greedy_BLEU	21.68	21.78	NaN	NaN	1
greedy_METEOR					

Figure 2: Metrics comparison — beam/greedy scores side by side (NaN = metric not logged in that run)

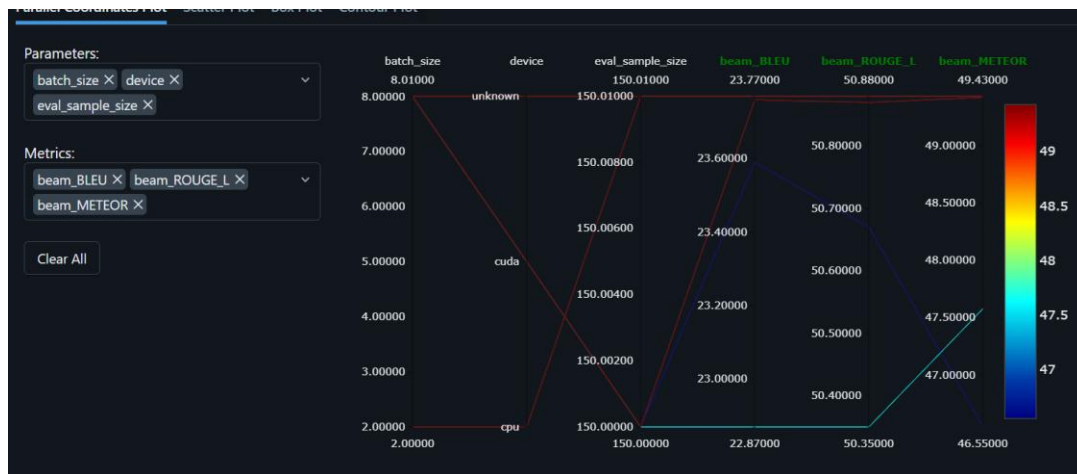


Figure 3: Parallel coordinates — parameters + beam metrics (color = beam_METEOR; red = best)

7. Next Steps

Scale fine-tuning to 1,000+ images with early stopping and validation-loss tracking, save the best checkpoint, and build a Gradio/Streamlit deployment interface for the final project deliverable.